

Setup & Initialization

1 Hardware Setup

- Connect the **LPC2148 ARM7** microcontroller.
- Connect the **16×2 LCD** for system information and status display.
- Connect the **4×4 Keypad** for user configuration.
- Connect the **RTC** for time and date management.
- Connect **LEDs and Buzzer** for hydration reminders.
- Connect the **Drink/Acknowledge Push Button** for recording water consumption.

2 Software Setup

- Install **Keil µVision** with ARM7/LPC2148 support.
- Create/open the project in Keil.
- Add all required source and header files:
 - **main.c**
 - **lcd.c** /
 - **lcd.h** **kpm.c**
 - / **kpm.h** **rtc.c**
 - / **rtc.h**
 - **hydration.c** /
 - **hydration.h** **reminder.c**
- Required delay, buzzer, and interrupt files.

3 Project Configuration

- Select the **LPC2148** target device.
- Configure the required **CPU clock frequency**.
- Configure GPIO pins for LCD, keypad, LEDs, buzzer, and push button.
- Initialize the **RTC** with the required time and date.
- Initialize the **LCD, keypad, interrupts, LEDs, and buzzer**.

4 Build & Program

1. Build the project in Keil.
2. Resolve any compilation or linker errors.
3. Generate the **.hex** file.
4. Load the HEX file into the LPC2148 using the required programmer/simulator.

5 System Startup

After reset, the system performs initialization and displays the welcome screen: